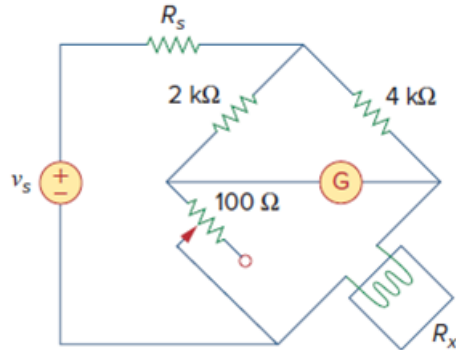


Problem

The Wheatstone bridge circuit shown in Fig. 4.146 is used to measure the resistance of a strain gauge. The adjustable resistor has a linear taper with a maximum value of $100\ \Omega$. If the resistance of the strain gauge is found to be $42.6\ \Omega$, what fraction of the full slider travel is the slider when the bridge is balanced?

Figure. 4.146:



Step-by-step solution

Step 1 of 2

Refer to Figure 4.146 in the textbook.

Write the formula for the unknown resistance in a Wheat-stone bridge in balanced condition.

$$R_x = \left(\frac{R_3}{R_1} \right) R_2$$

Step 2 of 2

Substitute $42.6\ \Omega$ for R_x , $4\ \text{k}\Omega$ for R_3 and $2\ \text{k}\Omega$ for R_1 .

$$42.6 = \frac{4}{2} R_2$$

$$R_2 = 21.3\ \Omega$$

R_2 is a variable resistor having maximum resistance of $100\ \Omega$.

Therefore, calculate the fraction of the full slider (maximum resistance) for balanced condition.

$$\begin{aligned} \alpha &= \frac{R_2}{R_{2\max}} \\ &= \frac{21.3}{100} \\ &= 0.213 \end{aligned}$$

Hence, the value of the fraction of the slider for balanced condition is 0.213.